

Homework 5: Statistical Phylogenetics

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Problem 12: Intuition on time-reversible Markov chains

1. Time-reversed process transition probabilities

For a time reversed-process:

$$\begin{aligned}\hat{T}_{kl}(s, t) &:= Pr(\hat{X}(s+t) = l | \hat{X}(s) = k) = P(\hat{X}(s+t) = l | \hat{X}(s) = k) \\ &= P(X(\tau - (s+t)) = l | X(\tau - s) = k)\end{aligned}$$

Applying the law of conditional probability:

$$\begin{aligned}&= \frac{P(X(\tau - s - t) = l, X(\tau - s) = k)}{P(X(\tau - s) = k)} = \frac{P(X(\tau - s - t) = l, X(\tau - s) = k)}{X(0)} = \\ &\quad \frac{P(X(\tau - s - t) = l, X(\tau - s) = k)}{\pi_k} =\end{aligned}$$

The numerator is the probability of reaching the state l at time $\tau - s - t$ (which is earlier than $\tau - s$) times the probability of reaching state k a time t later, given state l was reached beforehand:

$$\frac{P(X(\tau - s - t) = l) \cdot P(X(\tau - s) = k | X(\tau - s - t) = l)}{\pi_k} = \frac{\pi_l \cdot T_{lk}(t)}{\pi_k}$$

Therefore:

$$\rightarrow \hat{T}_{kl}(s, t) = \frac{\pi_l \cdot T_{lk}(t)}{\pi_k}$$

2. Original Markov chain reversability

Knowing

$$\hat{T}_{kl}(s, t) = \frac{\pi_l \cdot T_{lk}(t)}{\pi_k}$$

and

$$\hat{T}_{kl}(s, t) = \hat{T}_{kl}(t) \rightarrow \hat{T}_{kl}(t) = \frac{\pi_l \cdot T_{lk}(t)}{\pi_k}$$

Furthermore, since

$$\pi_k \cdot T_{kl}(t) = \pi_l \cdot T_{lk}(t)$$

for all $k, l \in [K], t \geq 0$, we can combine the two equations to obtain:

$$\hat{T}_{kl}(t) = \frac{(\pi_k \cdot T_{kl}(t))}{\pi_k} = T_{kl}(t) \rightarrow T_{kl}(t) = \hat{T}_{kl}(t)$$

Problem 13: Phylogenetic trees as Bayesian networks

1. Summation terms in brute force marginalisation

The joint probability $P(Y, Z|T)$ is as follows:

$$P(Y, Z|T) = P(Z_1) \cdot P(Y_4|Z_1) \cdot P(Z_2|Z_1) \cdot P(Y_3|Z_2) \cdot P(Z_3|Z_2) \cdot P(Y_2|Z_3) \cdot P(Y_1|Z_3)$$

Then, marginalising over Z to obtain $P(Y|T)$:

$$\begin{aligned} P(Y|T) &= \sum_Z (P(Z_1) \cdot P(Y_4|Z_1) \cdot P(Z_2|Z_1) \cdot P(Y_3|Z_2) \cdot P(Z_3|Z_2) \cdot P(Y_2|Z_3) \cdot P(Y_1|Z_3)) = \\ &= \sum_{Z_1, Z_2, Z_3} (P(Z_1) \cdot P(Y_4|Z_1) \cdot P(Z_2|Z_1) \cdot P(Y_3|Z_2) \cdot P(Z_3|Z_2) \cdot P(Y_2|Z_3) \cdot P(Y_1|Z_3)) \end{aligned}$$

Since each hidden variables, Z_1 , Z_2 and Z_3 , we are marginalising \$ has 4 states A, C, G, T , there are total of 64 summation terms

$$4^3 = 64$$

2. Summation terms after rearranging with Felsenstein's peeling algorithm.

Rearrangement with Felsenstein's peeling algorithm results as follows

$$\begin{aligned} P(Y|T) &= \sum_{Z_1, Z_2, Z_3} (P(Z_1) \cdot P(Y_4|Z_1) \cdot P(Z_2|Z_1) \cdot P(Y_3|Z_2) \cdot P(Z_3|Z_2) \cdot P(Y_2|Z_3) \cdot P(Y_1|Z_3)) \\ &= \sum_{Z_1} \left(P(Z_1) \cdot P(Y_4|Z_1) \sum_{Z_2} \left(P(Z_2|Z_1) \cdot P(Y_3|Z_2) \sum_{Z_3} (P(Z_3|Z_2) \cdot P(Y_2|Z_3) \cdot P(Y_1|Z_3)) \right) \right) \end{aligned}$$

The number of summation terms is:

$$(m - 1) * 4^2 + 4$$

where m is the number of internal nodes, or latent variables. Given there are three internal nodes, this results to be 36 summation terms:

$$(3 - 1) * 4^2 + 4 = 36$$

Significantly less compared to the naive 'brute force' marginalisation.

Problem 14 (data analysis): Learning phylogenetic trees from sequence alignment data

1. Loading alignment data

```
library(phangorn)
```

Warning: package 'phangorn' was built under R version 4.5.3

Loading required package: ape

Warning: package 'ape' was built under R version 4.5.3

```
library(ape)
```

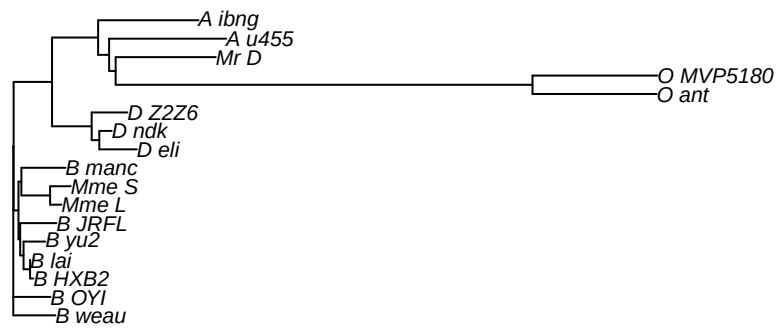
```
ParisRT <- read.dna("ParisRT.txt")
```

2. Topology tree with three nucleotide substitution models

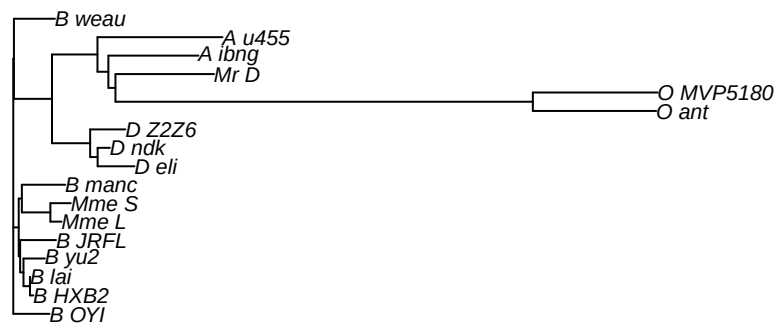
```
pd_K80 <- dist.dna(ParisRT, model = "K80")  
pd_TN93 <- dist.dna(ParisRT, model = "TN93")  
pd_JC69 <- dist.dna(ParisRT, model = "JC69")
```

```
joined_K80 <- NJ(pd_K80)  
joined_TN93 <- NJ(pd_TN93)  
joined_JC69 <- NJ(pd_JC69)
```

```
# cex to decrease the text size  
plot(joined_K80, cex=.7)
```



```
plot(joined_TN93, cex=.7)
```



```
plot(joined_JC69, cex=.7)
```



3. Log likelihood of Kimura nucleotide substitution model

```
phyData_K80 <- as.phyDat(ParisRT)
```

```
pml_K80 <- pml(joined_K80, phyData_K80, model= "K80")
```

```
logLik(pml_K80)
```

```
'log Lik.' -3003.487 (df=31)
```

```
print(pml_K80)
```

```
model: K80
```

```
loglikelihood: -3003.487
```

```
unconstrained loglikelihood: -2098.897
```

Rate matrix:

	a	c	g	t
a	0	1	1	1
c	1	0	1	1
g	1	1	0	1
t	1	1	1	0

Base frequencies:

	a	c	g	t
	0.25	0.25	0.25	0.25

The log-likelihood of the fitted Kimura model is -3003.486826.

4. Optimal parameters of Kimura nucleotide substitution model

```
optim1_pml_K80 <- optim.pml(pml_K80, optQ = TRUE, model = "K80")
```

```
optimize edge weights: -3003.487 --> -2992.981
optimize rate matrix: -2992.981 --> -2873.703
optimize edge weights: -2873.703 --> -2872.892
optimize rate matrix: -2872.892 --> -2872.881
optimize edge weights: -2872.881 --> -2872.881
optimize rate matrix: -2872.881 --> -2872.881
optimize edge weights: -2872.881 --> -2872.881
```

```
print(optim1_pml_K80)
```

```
model: K80
loglikelihood: -2872.881
unconstrained loglikelihood: -2098.897
```

Rate matrix:

	a	c	g	t
a	0.000000	1.000000	5.098082	1.000000
c	1.000000	0.000000	1.000000	5.098082
g	5.098082	1.000000	0.000000	1.000000
t	1.000000	5.098082	1.000000	0.000000

Base frequencies:

	a	c	g	t
	0.25	0.25	0.25	0.25

The values of the optimised rate matrix are:

-	a	c	g	t
a	0.000	1.000	5.098	1.000
c	1.000	0	1.000	5.098
g	5.098	1.000	0	1.000
t	1.000	5.098	1.000	0

Different, compared to those of the unoptimised model:

-	a	c	g	t
a	0.000	1.000	1.000	1.000
c	1.000	0	1.000	1.000
g	1.000	1.000	0	1.000
t	1.000	1.000	1.000	0

5. Optimisation of branch lengths, nucleotide substitution rates and tree topology; log likelihood of optimised model

```
optim2_pml_K80 <- optim.pml(pml_K80, optNni = TRUE, optEdge = TRUE, optQ = TRUE, model = "K80")
```

```
optimize edge weights: -3003.487 --> -2992.981
optimize rate matrix: -2992.981 --> -2873.703
optimize edge weights: -2873.703 --> -2872.892
optimize topology: -2872.892 --> -2859.775 NNI moves: 5
optimize rate matrix: -2859.775 --> -2859.682
optimize edge weights: -2859.682 --> -2859.681
optimize topology: -2859.681 --> -2859.681 NNI moves: 0
optimize rate matrix: -2859.681 --> -2859.681
optimize edge weights: -2859.681 --> -2859.681
```

```
print(optim2_pml_K80)
```

```
model: K80
loglikelihood: -2859.681
unconstrained loglikelihood: -2098.897
```

Rate matrix:

	a	c	g	t
a	0.000000	1.000000	5.262144	1.000000
c	1.000000	0.000000	1.000000	5.262144
g	5.262144	1.000000	0.000000	1.000000
t	1.000000	5.262144	1.000000	0.000000

Base frequencies:

a	c	g	t
0.25	0.25	0.25	0.25

The log likelihood of the optimised model is -2859.6809655, higher, indicating a better fit, compared to the firstly optimised model, -2872.8812889, which is higher (better fit) than the starting unoptimised model -3003.4868826.

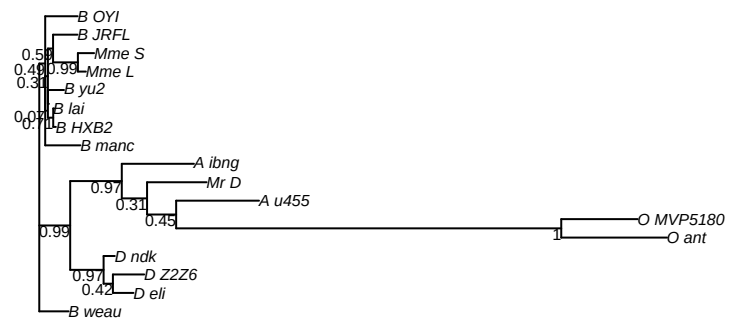
6. Bootstrapping

```
bs_pml_K80 <- bootstrap.pml(optim2_pml_K80, optNni = TRUE, model = "K80", control = pml.control)
```

It is bootstrapping with replacement the columns of the alignment of the species, or in other words, the nucleotide sites of the alignment.

7. Plotting and guilty nurses

```
plotBS(optim2_pml_K80$tree,bs_pml_K80,type = "phylogram",cex = 0.5)
```



The nurse “Mme S” is more likely to have effected “Mme L” because her HIV-virus strain sequence is phylogenetically more similar to that of patient “Mme L” compared to that of “Mr D” with “Mme L”.